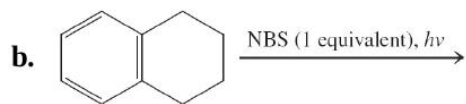
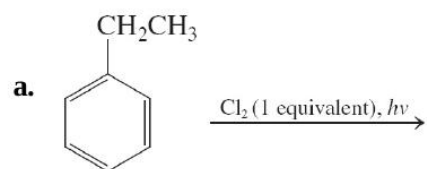


37. Give the expected major product(s) of each of the following reactions.



- 38.** Formulate a mechanism for the reaction described in [Problem 37](#), part (b).

39. Propose syntheses of each of the following compounds, beginning in each case with ethylbenzene. **(a)** (1-Chloroethyl)- benzene; **(b)** 2-phenylpropanoic acid; **(c)** 2-phenylethanol; **(d)** 2-phenyloxacyclopropane.

- 40.** Predict the order of relative stability of the three benzylic cations derived from chloromethylbenzene (benzyl chloride), 1-(chloromethyl)-4-methoxybenzene (4-methoxybenzyl chloride), and 1-(chloromethyl)-4-nitrobenzene (4-nitrobenzyl chloride). Rationalize your answer with the help of resonance structures.

41. By drawing appropriate resonance structures, illustrate why halogen atom attachment at the para position of the phenylmethyl (benzyl) radical is unfavored compared with attachment at the benzylic position.

CH 3

42. Triphenylmethyl radical, $(\text{C}_6\text{H}_5)_3\text{C}^\cdot$, is stable at room temperature in dilute solution in an inert solvent, and salts of triphenylmethyl cation, $(\text{C}_6\text{H}_5)_3\text{C}^+$, can be isolated as stable crystalline solids. Propose explanations for the unusual stabilities of these species.

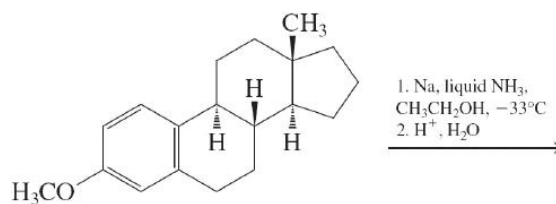
- 43.** Give the expected products of the following reactions or reaction sequences.

44. The hydrocarbon with the common name fluorene is acidic enough ($pK_a \approx 23$) to be a useful indicator in deprotonation reactions of compounds of greater acidity. Indicate the most acidic hydrogen(s) in fluorene. Draw resonance structures to explain the relative stability of its conjugate base.



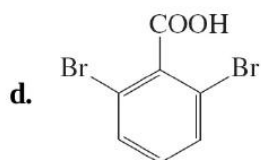
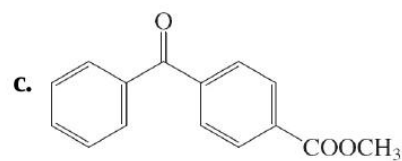
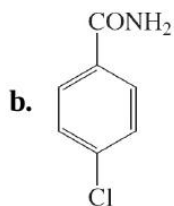
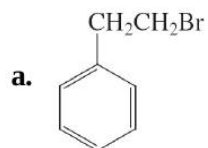
Fluorene

45. The Birch reduction shown below is widely used in the synthesis of a variety of highly biologically active steroids. Although the ^1H NMR spectrum of its product is complex, it exhibits a singlet at $\delta = 5.73$ ppm integrating to one hydrogen. In addition, the IR spectrum shows absorptions at $\tilde{\nu} = 1620$ and 1660 cm^{-1} , of medium and very strong intensities, respectively.



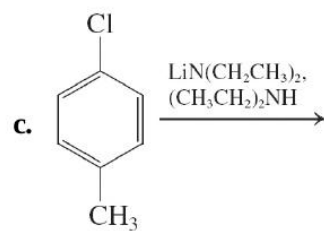
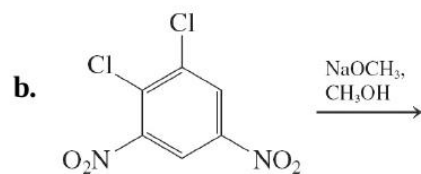
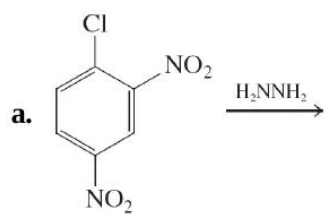
Propose a structure for the product and rationalize its formation

46. Outline a straightforward, practical, and efficient synthesis of each of the following compounds. Start with benzene or methylbenzene. Assume that the para isomer (but not the ortho isomer) may be separated efficiently from any mixtures of ortho and para substitution products. Work backward from the target.



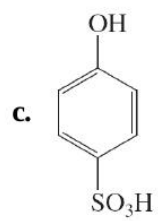
47. Rank the following molecules in descending order of reactivity toward hydroxide ion.

48. Predict the main product(s) of the following reactions. In each case, describe the mechanism(s) in operation.

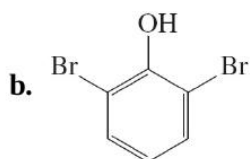
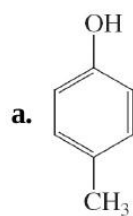


- 49.** Starting with aniline, propose a synthesis of akomide, an agent used to treat certain exotic fungal and protozoal infections in veterinary medicine. Several intermediates are shown to give you the general route.

57. Rank the following compounds in order of descending acidity.

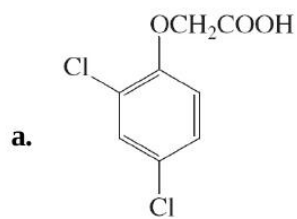


58. Design a synthesis of each of the following phenols, starting with either benzene or any monosubstituted benzene derivative.

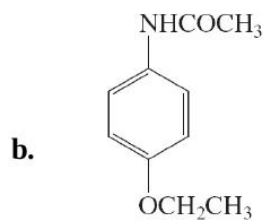


- c. The three benzenediols

59. Starting with benzene, propose syntheses of each of the following phenol derivatives.

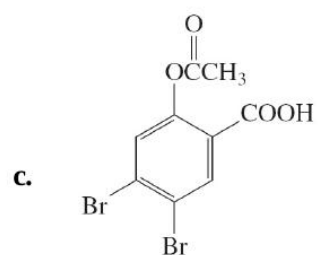


The herbicide 2,4-D



Phenacetin

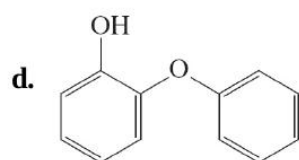
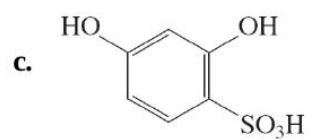
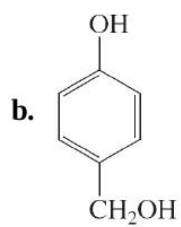
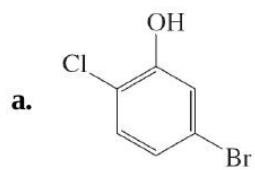
(A discontinued analgesic)



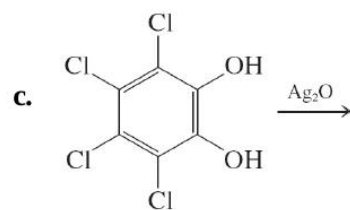
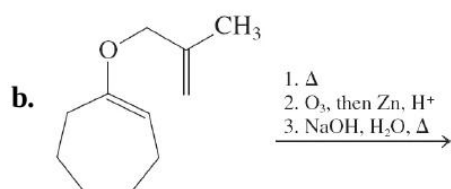
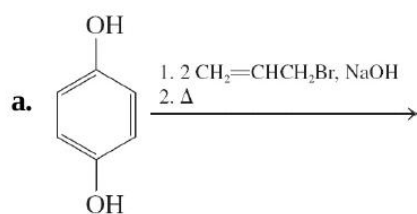
Dibromoaspirin

(An experimental drug for the treatment of sickle-cell anemia)

60. Name each of the following structures.



61. Give the expected product(s) of each of the following reaction sequences.



66. As mentioned in [Section 22-10](#), the Sandmeyer reactions, which entail the substitution of the nitrogen in arenediazonium salts by Cl, Br, CN , require copper(I) ion as a catalyst and proceed by complex radical mechanisms. Why do such substitutions not follow an $\text{S}_{\text{N}}2$ pathway? Why is the $\text{S}_{\text{N}}1$ mechanism, which operates in the corresponding displacements with OH^- and I^- , not effective?.

67. Formulate a detailed mechanism for the diazotization of aniline in the presence of HCl and NaNO₂. Then suggest a plausible mechanism for its subsequent conversion into iodobenzene by treatment with aqueous iodide ion (e.g., from K⁺I⁻). Bear in mind your answer to [Problem 66](#).

68. Show how you would convert 3-methylaniline into each of the following compounds: **(a)** methylbenzene; **(b)** 1-bromo- 3-methylbenzene; **(c)** 3-methylphenol; **(d)** 3-methylbenzonitrile; **(e)** *NN*-ethyl-3-methylaniline.

69. Devise a synthesis of each of the following substituted benzene derivatives, starting from benzene.

